

§16. Hilbert functions

prop. If $0 \rightarrow K \rightarrow N \rightarrow U \rightarrow 0$ exact seq of graded f.g. R -mod, then

$$\text{Hilb}_N(t) = \text{Hilb}_K(t) + \text{Hilb}_U(t)$$

Theorem 16.2. (Hilbert) Let $\mathbf{F}$ be a graded free resolution of a finitely generated graded R -module U . Write

$$F_i = \bigoplus_{p \in \mathbb{Z}} R^{c_{i,p}}(-p).$$

For each p suppose that $c_{i,p} = 0$ for $i \gg 0$. Then

$$\text{Hilb}_U(t) = \text{Hilb}_R(t) \sum_{i \geq 0} \sum_{p \in \mathbb{Z}} (-1)^i c_{i,p} t^p.$$

If $R = S$, then

$$\text{Hilb}_U(t) = \frac{\sum_{i \geq 0} \sum_{p \in \mathbb{Z}} (-1)^i c_{i,p} t^p}{(1-t)^n}.$$

The following cases are the main two cases when for each p we have $c_{i,p} = 0$ for $i \gg 0$, and Theorem 16.2 can be applied.

(1) The resolution $\mathbf{F}$ is finite.

(2) The resolution $\mathbf{F}$ is minimal. In this case $c_{i,p} = b_{i,p}^R(U)$ and we apply Proposition 12.3.

pf. 1. $\mathbf{F}$ is finite

2. In general

1. By induction on the length of $\mathbf{F}$.

$$\textcircled{1} \quad 0 \rightarrow K \rightarrow F_0 \rightarrow U \rightarrow 0$$

$$\text{Hilb}_U(t) = \text{Hilb}_{F_0}(t) - \text{Hilb}_K(t).$$

$$F_0 = \bigoplus_p R(-p)^{c_{0,p}},$$

$$\text{Hilb}_{F_0}(t) = \text{Hilb}_R(t) \sum_{p \in \mathbb{Z}} c_{0,p} t^p.$$

Since the resolution of K is

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow 0$$

By induction hypothesis

$$\text{Hilb}_k(t) = \text{Hilb}_k(t) \sum_{i \geq 1} \sum_{p \in \mathbb{Z}} (-1)^{i-1} c_{i,p} t^p.$$

$$\Rightarrow \text{Hilb}_0(t) = \dots$$

2. In general
for each p , $c_{i,p} = 0$ for $i \gg 0$. $\sum \sum$

Q.

Ex 16.4 $f_1, \dots, f_q$ S -regular with degree $a_1, \dots, a_q$

$$\text{Hilb}_{S/(f_1, \dots, f_q)}(t) = \frac{\prod_{i=1}^q (1-t^{a_i})}{(1-t)^n}$$

pf. Koszul complex $K(f_1, \dots, f_q; S)$

$$0 \rightarrow \underbrace{S(-a_1 - \dots - a_q)}_{F_q} \rightarrow \dots \rightarrow \underbrace{\bigoplus_{i=1}^q S(-a_i)}_{F_1} \rightarrow S \rightarrow S/(f_1, \dots, f_q) \rightarrow 0$$

$F_i = \bigoplus_{1 \leq j_1 < \dots < j_i \leq n} S(-a_{j_1} - \dots - a_{j_i})$

Q

Corollary 16.5. Let V be a graded finitely generated S -module. There exists a polynomial $g(i) \in \mathbb{Q}[i]$ of degree $< n$ such that $g(j) = \dim_k(V_j)$ for all $j \gg 0$.

The polynomial $g(i)$ in Corollary 16.5 is called the **Hilbert polynomial** of V . The following theorem can be proved by induction on the dimension, cf. [Bruns-Herzog, Theorem 4.1.3].

pf. $\text{Hilb}_0(t) = \frac{\sum_{i \geq 0} \sum_{p \in \mathbb{Z}} (-1)^i c_{i,p} t^p}{(1-t)^n} = \sum_p d_p t^p$

$$\frac{1}{(1-t)^n} = \sum_{k \geq 0} \binom{n-1+k}{n-1} t^k$$

$$\begin{aligned}
\text{Hilb}_V(t) &= \left(\sum_p d_p t^p \right) \left(\sum_{k=0}^{\infty} \binom{n-1+k}{n-1} t^k \right) \\
&= \sum_{k=0}^{\infty} \sum_p d_p \binom{n-1+k}{n-1} t^{p+k} \\
&= \sum_j \dim_K(V_j) t^j \\
\text{for } j \geq 0, \quad \dim_K(V_j) &= \sum_p d_p \binom{n-1+j-p}{n-1} \\
&\quad \text{is a polynomial with degree } < n.
\end{aligned}$$

$F_i = \bigoplus_p R(-p)^{c_{i,p}}$

□

Theorem 16.6. Let V be a graded finitely generated S -module. Its Hilbert polynomial has degree $\dim(V) - 1$.

Euler polynomial

V graded f.g. S -mod TF finite

$$E_F(t) = \sum_{i \geq 0} \sum_{p \in \mathbb{Z}} (-1)^i c_{i,p} t^p \in \mathbb{Z}[t]$$

$E_F(t)$ doesn't depend on choice of resolution

$$E_F(1) = \sum_{i \geq 0} (-1)^i c_i$$

TF minimal, $E_V(1) = \sum_{i \geq 0} (-1)^i b_i^S(V)$

$$\text{Hilb}_V(t) = \frac{E_F(t)}{(1-t)^n}$$

↑

Set $h(t) = \frac{E_F(t)}{(1-t)^q}$, where q is the maximal power such that $(1-t)^q$ divides $E_F(t)$. Set $s = n - q$. Thus, $h(1) \neq 0$ and

$$\text{Hilb}_V(t) = \frac{h(t)}{(1-t)^s}.$$

The number $h(1)$ is called the **multiplicity** (or the **degree**) of the module V , and is denoted $\text{mult}(V)$.

Theorem 16.7. Let V be a graded finitely generated S -module. If V is artinian, then $\text{Hilb}_V(t) = h(t)$ and $h(1)$ is its length. If V is not artinian, then (using the preceding notation) we have

(1) $\text{Hilb}_V(t) = \frac{h(t)}{(1-t)^{\dim(V)}} = s \quad \checkmark$

$h(t) = \sum_i \dim(V_i) t^i$

(2) The leading coefficient of the Hilbert polynomial $g(i)$ is

$= \text{length}(V)$

$\frac{h(1)}{(\dim(V) - 1)!} \quad \checkmark$

(3) If $h(t) = h_r t^r + \dots + h_1 t + h_0$, then

$g(i) = \sum_{0 \leq j \leq r} h_j \binom{\dim(V) - 1 + i - j}{\dim(V) - 1} \quad \checkmark$

(4) $\dim_k(V_i) = g(i)$ for $i \geq \deg(h(t))$. $\checkmark$

We use the convention $\binom{p}{0} = 1$ for every p .

If we can compute the graded Betti numbers of V , then we can compute the following invariants:

- the Hilbert series of V ; by Theorem 16.2
- the dimension of V ; by Theorem 16.7(1)
- the Hilbert polynomial of V ; by Theorem 16.7(4)
- the multiplicity of V ; by Theorem 16.7(3).

pf. $\text{Hilb}_V(t) = \frac{h(t)}{(1-t)^s} \quad s \leq n$

$h(t) = h_r t^r + h_{r-1} t^{r-1} + \dots + h_0$

$\forall s \geq 1$

$\frac{1}{(1-t)^s} = \sum_{i \geq 0} \binom{s-1+i}{s-1} t^i$

$\text{Hilb}_V(t) = \sum_{j=0}^{\infty} \dim_k(V_j) t^j$

$\Rightarrow g(i) = \sum_{0 \leq j \leq r} h_j \binom{s-1+i-j}{s-1}$

$\deg(g(i)) = s-1$

Thm 16.6 $\Rightarrow \dim(V) - 1 = s - 1 \Rightarrow \underline{\dim V = s} \geq 1$

(1) (2) (3) (4) $\checkmark$.

② $s < 1$, $s \leq 0 \Rightarrow$ artinian

$$\begin{aligned} \text{Hilb}_V(t) &= h(t) (1-t)^{-s} \\ &= h(t) \left(\sum_{0 \leq i \leq -s} \binom{-s}{i} (-1)^i t^i \right) \end{aligned}$$

$g(i) = 0$ $\deg(g(i)) = -1$.
 Thm 16.6 $\Rightarrow \dim V = 0 \Rightarrow V$ artinian.

Exercise 16.9. Let $f_1, \dots, f_q$ be a homogeneous regular sequence of forms of degrees $a_1, \dots, a_q$. Compute the Hilbert polynomial of $S/(f_1, \dots, f_q)$ and show that $\text{mult}(S/(f_1, \dots, f_q)) = a_1 \dots a_q$.